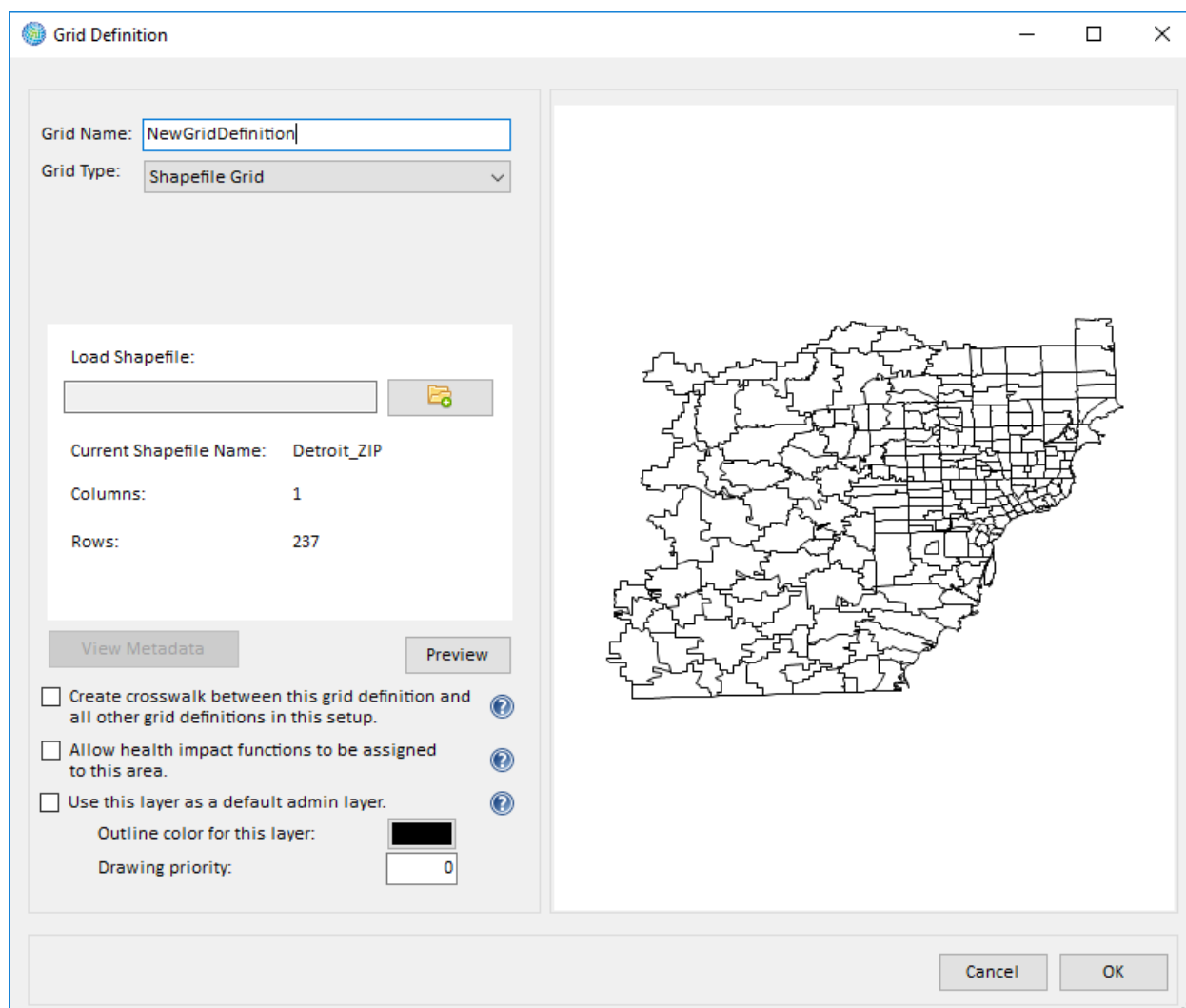




When you are satisfied that the shapefile looks correct and have checked the appropriate boxes on the lower left, click **OK**. This will save the shapefile and bring you back to **Manage Grid Definitions** window.¹⁴ Note that the **Grid Type** box displays the type of grid for each of your grid definitions. The **View Metadata** button will allow you to see and edit any comments that were previously entered.

Click **OK** when you are finished loading grid definitions. The **Modify Datasets** screen will now list the **Grid Definitions** that you have just created. At any time, you may click the **Manage** button to add, modify, or delete grid definitions.

Warning

If you delete a **Grid Definition**, you will also permanently delete any gridded data (e.g., **Population** or **Incidence**) that is dependent on it.

¹⁴ Shapefiles are saved to C:\Users\\AppData\Local\BenMAP-CE\Data\Shapefiles\

WARNING! If you delete a **Grid Definition**, you will permanently delete any gridded data that is dependent on it, such as any **Incidence/Prevalence, Population, and Variable Datasets** that use this particular **Grid Definition**. As we discuss each of these other setup elements below, we will describe how this might happen.

4.1.2 Pollutants

The **Pollutants** section of a setup specifies the pollutants that BenMAP-CE will analyze and defines the air quality metrics to be used by BenMAP-CE. You are not importing air pollution data, but rather naming your pollutants and defining the measures or metrics BenMAP-CE will use when performing an analysis for each pollutant. You may include any pollutant, though typically air pollutants such as particulate matter, ozone, sulfur dioxide, and carbon monoxide are used in a BenMAP-CE analysis.

A key concept for pollutants is the **Metric**. Air quality metric describes the period of the day over which the pollutant observations are averaged. For example, a metric of *D24HourMean* is a daily average of hourly measurements. A metric of *D8HourMax* is the average of the 8-hour period during the day when pollutant levels are the highest (see Table 4-2 below). The air quality change must be expressed in a metric that matches the metric used by the health impact function; this concept is discussed further below.

Fundamental Concept – Pollutant Metric

The **metric** expresses the time period over which air quality values are modeled or observed and the type of statistic being used (e.g., average, maximum or minimum). For example, the metric **D24HourMean** represents the average concentration for the sampling day. **D8HourMax** represents the highest of the 8-hour moving averages during the day. One pollutant may have multiple metrics defined. In order to use a Health Impact Function, the BenMAP pollutant definition must include the metric associated with that function.

In general, air pollution data in BenMAP-CE is hierarchical – a pollutant can have multiple **Metrics**, each of which has multiple **Statistics** (these are automatically calculated by BenMAP-CE) and which can have multiple **Seasonal Metrics**. Similarly, **Seasonal Metrics** have multiple **Statistics**. Furthermore, air pollution data can be provided to BenMAP-CE at any of these levels, in addition to the daily and hourly observation level, as described in more detail in Section 4.3.

Table 4-2. Example Calculation of D8HourMax

Hourly Period ¹	Hourly average O ₃ concentration ² (ppm)	Moving 8-hour average ³	D8HourMax ⁴
07:00	0.000	---	← 0.049
08:00	0.005	---	
09:00	0.010	---	
10:00	0.015	---	
11:00	0.020	---	
12:00	0.025	---	
13:00	0.030	---	
14:00	0.035	---	
15:00	0.040	0.020	
16:00	0.045	0.025	
17:00	0.050	0.030	
18:00	0.055	0.035	
19:00	0.060	0.040	
20:00	0.055	0.044	
21:00	0.050	0.047	
22:00	0.045	0.048	
23:00	0.040	<u>0.049</u>	
00:00	0.035	0.048	
01:00	0.030	0.047	
02:00	0.025	0.044	
03:00	0.020	0.040	
04:00	0.015	0.035	
05:00	0.010	0.030	
06:00	0.000	0.024	

¹ Days for measuring ozone start and end at 7:00 AM local standard time.

² Hourly average is the average of individual measurements taken during the hour.

³ Moving 8-hour average is the average of the hour and the proceeding 7 hours.

⁴ *D8HourMax* is the maximum of the moving 8-hour averages.

Reference: Federal Register 79 FR 75233, Dec. 17, 2004.

<https://www.federalregister.gov/articles/2014/12/17/2014-28674/national-ambient-air-quality-standards-for-ozone#h-193>

Add a Pollutant

Air pollution data in BenMAP-CE is of two types: (1) point source monitoring data and (2) **Grid Definition**-based modeling data. For both types, the data must be associated with a particular pollutant. Table 4-3 describes these variables used to define a pollutant in BenMAP-CE.

Table 4-3. BenMAP-CE Pollutant Definitions

Pollutant Field Name	Notes
Pollutant ID	Unique name for the pollutant which will be referenced in health impact functions, associated with monitoring and modeling data, etc.
Observation Type	Pollutants may have hourly observations or daily observations. In the United States, Ozone has hourly observations, while PM ₁₀ and PM _{2.5} have daily observations.
Metrics	Daily values calculated directly from daily observations, or through various mathematical manipulations of hourly observations. Typical ozone metrics include the highest hourly observations during the course of each day, the mean of all twenty four hourly observations, etc.
Seasonal Metrics	Seasonal values calculated from metric values. In the United States, for example, quarterly means are calculated for PM _{2.5} from daily means.

To add pollutant definitions to BenMAP-CE, click **Modify Datasets** on the menu bar. Then click on the **Manage** button below the **Pollutants** group. The **Manage Pollutants** window will appear. Here you may click **Add** to add a new **Pollutant**, **Delete** to remove a previously defined pollutant, or **Edit** to modify an existing pollutant.

